Kinetics in Organic Chemistry

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You have had many courses that featured detailed analysis of reaction kinetics and thermodynamics. In this meeting we will explore the application of kinetics to investigating reaction mechanism.

Any single step in a reaction mechanism could be described as first-order or second-order reaction. The rate laws of the steps in a mechanism may be combined to result in rate laws that describe higher reaction orders and even non-integer orders. If the kinetics of a reaction fit the mathematical model of the rate law then the proposed model might be correct. If they do not fit then the model is wrong. Being able to propose a rate law and fit the observed data to the model is an important tool in physical organic chemistry.[1]

[1] Read chapter 7 of the textbook.

Complete Before Class

1. Write out single step reactions that can be described as first and second order. Propose a rate law for each and integrate the rate law to obtain a model that describes the concentration of reactants and products over time.

2. The data in table 1 describes the decrease in reactant concentration over time. Determine the order of reaction and the rate constant.[2]

Table 1: Concentration vs. time data for a reactions.

time (/min)	Reactant (/M)
0	0.250
0.042	0.213
0.092	0.175
0.119	0.156
0.151	0.138
0.280	0.081
0.435	0.044

[2] Ch. 7.4

3. For a reaction of the form $n\,\mathrm{A} \longrightarrow \mathrm{B}$, show that a plot of $log(\nu_{t=0})$ vs. $[A]$ will have a slope equal to the order of the reaction, n.[3]

[3] $\nu_{t=0}$ is the rate at $t = 0$. This is an example of the initial rate method.

4. The hydrolysis of tButyl bromide is slower when excess concentrations of NaBr are present. Use the steady-state method to derive a rate law for this reaction that explains the observation.[4]

[4] Ch. 7.5

5. State the Hammond postulate.[5]

[5] Ch. 7.3

6. Describe the principle of microscopic reversibility.[5]

7. Define the Curtin-Hammett principle.[5]

Textbook Questions

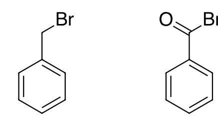
As you review your previous experience in kinetics try these problems:
Ch. 7: 3–5, 9, 11, 13, 14

For the Class Meeting

1. Consider the hydrolysis of benzyl bromide.
   (a) Draw out a diagram of the energy vs. the reaction coordinate.[6]
   (b) Identify all transition states and intermediates in the diagram.[7]
   (c) Identify the rate-determining step.
   (d) Draw diagrams representing the 3D structure of all molecules and transition states along the reaction coordinate. Justify your structures for the transition states by using the **Hammond postulate**.

[6] Ch. 7.1

[7] Ch. 7.2



BnCl BzCl

Figure 1: Benzyl bromide (BnCl) and benzoyl bromide (BzCl)

2. Now reconsider the hydrolysis of benzyl bromide. There are another other possible mechanism. Repeat the above analysis with this alternate mechanism.

3. Use a More O'Ferrall-Jencks plot to describe both reaction coordinates from problems 1 and 2 above.[8]
   (a) Locate the transition states and/or intermediates on this plot for both mechanisms.
   (b) Identify each pathway as either S_N1 or S_N2.
   (c) For the S_N2 pathway, use the More O'Ferrall-Jencks plot to describe how the structure of the transition state would change in each of the following cases.
      i. We use nitrobenzyl bromide rather than benzyl bromide.[9]
      ii. We use methoxybenzyl bromide rather than benzyl bromide.[10]
      iii. We use benzyl chloride rather than benzyl bromide.[11]
      iv. We use a better nucleophile than water. Perhaps hydroxide?

[8] Ch. 7.8

[9] The nitro group is electron-withdrawing.

[10] The methoxy group is electron-donating.

[11] The choride group is not as good a leaving group as bromide.

4. Alkyl bromides can eliminate via an E1 or E2 mechanism. Consider the observed preference for anti-elimination in alkyl bromides. Use the **principal of microscopic reversibility** to show that the mechanism for addition of HBr to an alkene is unlikely to occur via a concerted process.

5. (2R,3R)-2-Bromo-3-methylpentane undergoes E2 elimination to give only the (E)-3-Methyl-2-pentene product. Use this reaction as an example of the **Curtin-Hammett principle**.

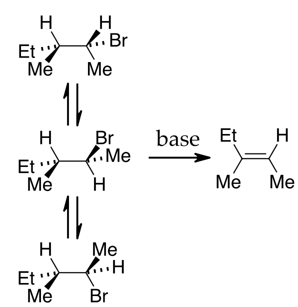


Figure 2: E2 elimination in (2R,3R)-2-bromo-3-methylpentane

6. Explain how the regiochemistry of an enolate addition can be manipulated through **kinetic and thermodynamic control**.[5]

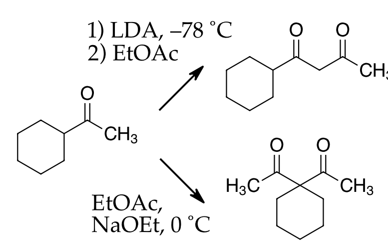


Figure 3: regiochemical control in an enolate reaction with an ester.

More Textbook Questions

After today's class discussion try these problems: Ch. 7: 8, 10, 12, 15–17, 23–25

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will…

- have reviewed the basics of reaction kinetics and rate laws.
- be able to derive a rate law for single-step and multi-step reactions.
- be able to interpret reaction kinetics data and obtain rate constants by fitting data to a mathematical model of a rate law or integrated rate law.
- be able to use the Arrhenius equation and the Eyring equation to determine activation parameters for a reaction and predict reaction rates are a given temperature.
- be able to define and interpret the consequences of the Hammond postulate, the principle of microscopic reversibility and the Curtin-Hammett principle.
- be able to draw reaction coordinate diagrams for a reaction and locate all reactants, intermediates, products and transition states.
- be able to predict the structure of a transition state and how changing the structure of the reactant can change the energy and the structure of the transition state.
- be able to use the More O'Ferrall-Jencks plot as a tool to accomplish the above.
- **Computer Skill**: be able to use an interactive *Python* notebook to perform and document line fits of experimental data.

A Note on More O'Farrell–Jencks Plots

The More O'Ferrall-Jencks plot[12] enables chemists to better interpret changes in transition state structure. Let us explore an example.

[12] Read ch. 7.8

Many reactions have two things happening at once in the RDS. In a typical S_N2 reaction we have the leaving group leaving and the nucleophile forming a new bond. One distance is getting longer and the other shorter as the reaction proceeds. The reaction coordinate can be plotted in three dimensions using the system popularized by Rory More O'Ferrall[13] and William Jencks.[14]

These plots are especially useful[15] in interpreting the correlation of structure with reactivity and we will be using them many more times. So let us explore a simple example using an elimination reaction.

[13] "Rory A. More O'Ferrall" S.J. Ainsworth, *Chem. Eng. News*, **2012**, *90*, https://cen.acs.org/articles/90/i50/RoryOFerrall.html

[14] William P. Jencks, 1927-2007" J.F. Kirsch, J.P. Richard, *Biog. Mem. Natl. Acad. Sci.*, **2010**, 3-20 http://www.nasonline.org/publications/biographical-memoirs/memoir-pdfs/jencks-william.pdf

The Example of Elimination

An elimination reaction[16] can be described by two changes. We move a proton from the alkyl halide to a base and the leaving group departs. We could measure the former by the distance between the base and the proton; it would start at VDW distance and move toward a bond distance. We could measure the latter simply by the bond distance to the leaving group; it would change from bond distance to VDW as shown in figure 4.

[15] For a modern example see "Mapping transition state structures for thiophosphinoyl group transfer between oxyanionic nucleophiles in water and aqueous ethanol solvents", G.I. Kalu, C.I. Ubochi, I. Onyido, *New J. Chem.*, **2022**, *46*, 12981-12993. https://doi.org/10.1039/D2NJ02008D

[16] Ch. 10.13

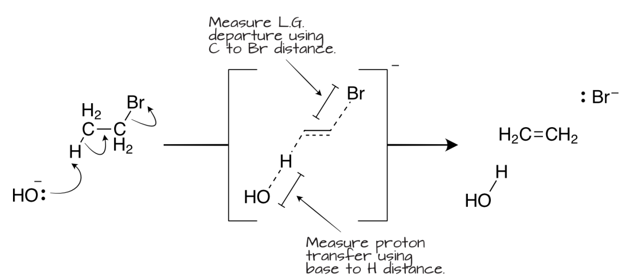


Figure 4: The E2 elimination mechanism with ethyl bromide

One change could be complete before the other begins (the E1 mechanism) or both changes could be simultaneous (the E2 mechanism). The beauty of the More O'Ferrall-Jencks plot is that we can imagine everything in-between. For example we could have a concerted E2 mechanism that has significant departure of the leaving group (and a developing positive change) before the base makes much progress in accepting the proton. This difference in T.S. structure can be elucidated using the methods of physical organic chemistry that we will begin exploring soon. With this method we can also visualize the less common E1cb mechanism for elimination,[16]

Consider the plots on the following page and think about the exact structure of each transition state.

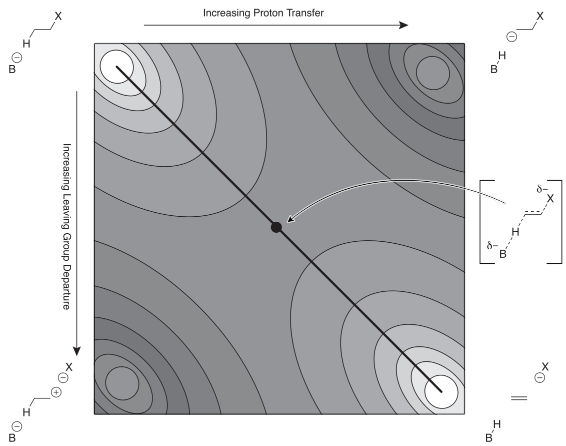


Figure 5: A More O'Ferrall-Jencks plot describing the E2 elimination mechanism

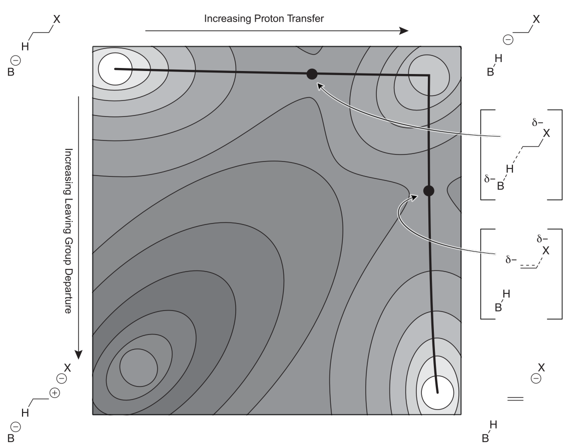


Figure 6: A More O'Ferrall-Jencks plot describing the E1cb elimination mechanism

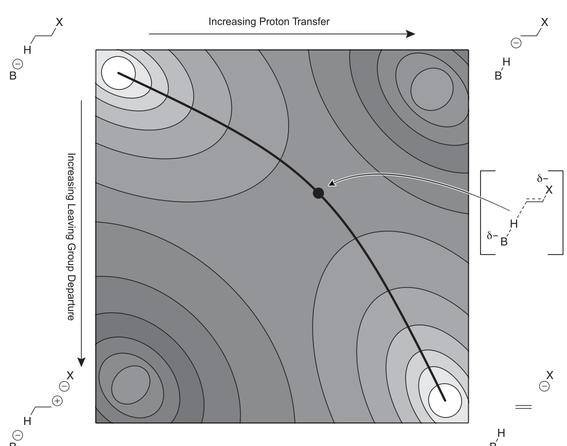


Figure 7: A More O'Ferrall-Jencks plot describing an asymmetrical E2 elimination mechanism

What path would you draw in a More O'Ferrall-Jencks plot that describes the E1 mechanism?

A Note on the Arrhenius Equation

You have been told that when using the Arrhenius equation the log A value is less that 13.2 when the entropy of activation is negative and greater than 13.2 when it is positive.[17] What is the justification for that rule-of-thumb? This discussion will explore that idea.

[17] See ch. 7.2

The Arrhenius equation[18] relates the temperate to the rate of a reaction using the empirical values A and E_a. The latter can be considered a measure of "activation energy" and the former is a fudge-factor that accounts for all manner of other aspects of the reaction, including the entropy of activation. This equation assumes no knowledge of the process (it was created before chemists knew about transition states) and is often used to predict reaction rates at various temperatures. But, it only gives hints about the enthalpy and entropy changes in the process.

[18] The Arrhenius equation

$$k = A \cdot e^{\frac{-E_a}{RT}}$$

and the linear form

$$\ln k = \ln A - \frac{E_a}{RT}$$

The Eyring equation[19] is used for single steps in a reaction. If you can determine the rate constants for a single-step in a reaction *vs.* temperature then you can determine the values for the enthalpy and entropy of activation for the single transition state in that single step.

[19] The Eyring equation

$$k = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^{\ddagger}}{R}} \cdot e^{\frac{-\Delta H^{\ddagger}}{RT}}$$

and the linear form

$$\ln k = \ln\left(\kappa \frac{k_B}{h}\right) + lnT + \frac{\Delta S^{\ddagger}}{R} - \frac{\Delta H^{\ddagger}}{RT}$$

They look the same, don't they. Both include a pre-exponential term and exponential terms. For a simple one-step reaction we could state that the Arrhenius plot and the Erying plot must result in the same answer at a given temperature and the two plots must have the same slope (both plots must give the same line). Therefore, the derivative w.r.t. T (the x-axis) in both linear plots must be the same.

$$k = A \cdot e^{\frac{-E_a}{RT}} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^{\ddagger}}{R}} \cdot e^{\frac{-\Delta H^{\ddagger}}{RT}}$$

Equate the Arrhenius and the Erying equations.

Some calculus review for you...

$$\frac{\partial}{\partial x} c = 0$$

$$\frac{\partial}{\partial x} cx = c \frac{\partial}{\partial x} x$$

$$\frac{\partial}{\partial x} \ln x = \frac{1}{x}$$

$$\frac{\partial}{\partial x} x^n = nx^{n-1}$$

$$\frac{\partial}{\partial x} \frac{1}{x} = -\frac{1}{x^2}$$

$$\ln k = \ln A - \frac{E_a}{RT} = \ln\left(\kappa \frac{k_B}{h}\right) + lnT + \frac{\Delta S^{\ddagger}}{R} - \frac{\Delta H^{\ddagger}}{RT}$$

$$\frac{\partial}{\partial T} \ln A - \frac{\partial}{\partial T} \frac{E_a}{RT} = \frac{\partial}{\partial T} \ln\left(\kappa \frac{k_B}{h}\right) + \frac{\partial}{\partial T} lnT + \frac{\partial}{\partial T} \frac{\Delta S^{\ddagger}}{R} - \frac{\partial}{\partial T} \frac{\Delta H^{\ddagger}}{RT}$$

$$0 + \frac{E_a}{RT^2} = 0 + \frac{1}{T} + 0 + \frac{\Delta H^{\ddagger}}{RT^2}$$

$$\frac{E_a}{RT^2} = \frac{1}{T} + \frac{\Delta H^{\ddagger}}{RT^2}$$

$$E_a = RT + \Delta H^{\ddagger}$$

multiplied both sides by RT^2

We now have a term for E_a in single-step simple reactions that can be used to make the Arrhenius equation the same as the Eyring equation.

$$A \cdot e^{\frac{-\left(RT+\Delta H^{\ddagger}\right)}{RT}} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^{\ddagger}}{R}} \cdot e^{\frac{-\Delta H^{\ddagger}}{RT}}$$

Some math review...

$$e^{a+b} = e^a e^b$$

$$A \cdot e^{-\frac{RT}{RT}} e^{\frac{-\Delta H^{\ddagger}}{RT}} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^{\ddagger}}{R}} \cdot e^{\frac{-\Delta H^{\ddagger}}{RT}}$$

$e^{\frac{-\Delta H^{\ddagger}}{RT}}$ is on both sides and cancels

$$A \cdot e^{-1} = \kappa \frac{k_B}{h} T \cdot e^{\frac{\Delta S^{\ddagger}}{R}}$$

$$A = \kappa \frac{k_B}{h} e \cdot T \cdot e^{\frac{\Delta S^{\ddagger}}{R}}$$

multiplied both sides by e

We now have a term for the Arrhenius pre-exponential value, A, in terms of $\Delta S^{\ddagger}$. Let us calculate the value of $\log A$ and see how this can lead us to that rule of thumb.

More math review...

$$\ln e^x = x$$

$$\log e^x = x \log e$$

$$\log e = \frac{1}{2.303}$$

$$\log A = \log \kappa \frac{k_B}{h} e + \log T + \frac{\Delta S^{\ddagger}}{R} \cdot \log e$$

If we set $\Delta S^{\ddagger} = 0$ and work at room temperature, we see that...

$$\log A = \log \left(1 \cdot \frac{1.381 \times 1^{-23}}{6.626 \times 10^{-34}} \cdot 2.7183\right) + \log 298 + \frac{0}{8.314 \times 2.303} = 13.2$$

Euler's number	$e = 2.7183$
Boltzmann const.	$k_b = 1.381 \times 1^{-23}$ J/K
Planck's const.	$h = 6.626 \times 10^{-34}$ J/s
Transmission coeff.	$\kappa = 1$
Gas const.	$R = 8.314$ J/mole·K

At room temperature, we observe that values of $\log A$ that are less than 13.2 would have a negative entropy of activation and values that are more than 13.2 would have a positive entropy of activation. This dividing line is temperature-dependant but does not change much over usual laboratory temperatures (e.g., $\log 298 = 2.5$ and $\log 600 = 2.8$).

A rule of thumb is not a law. This chemical aphorism provides a quick way to evaluate the nature of the transition state but always remember that it may not give the correct answer.[20] It is best used for simple first and second order reactions in solution. With those caveats in mind, here is the official rule of thumb:

[20] For more information on this topic see "The 'Yard Stick' to Interpret the Entropy of Activation in Chemical Kinetics: A Physical-Organic Chemistry Exercise", R. Sanjeev, D. A. Padmavathi, V. Jagannadham, *World Journal of Chemical Education*, **2018**, *6*, 78–81. https://doi.org/10.12691/wjce-6-1-12

> For uncomplicated reactions in solution, we can estimate the $\Delta S^{\ddagger}$ using the following rule. If $\log A > 10.8 + \log T$ then the $\Delta S^{\ddagger}$ is positive; If $\log A < 10.8 + \log T$ then the $\Delta S^{\ddagger}$ is negative.

We can solve for $\Delta S^{\ddagger}$ if we wish and then see how different $\log A$ values reveal the value of the entropy of activation.

$$\Delta S^{\ddagger} = \frac{R}{\log e}\left(\log A - \log \kappa \frac{k_B}{h} e - \log T\right)$$

The value of $\Delta S^{\ddagger}$ increases in magnitude by $2.3R$ (approx. 20 J/mole·K) for each unit we are away from the rule of thumb value for $\log A$ (table 2.)

Table 2: $\Delta S^{\ddagger}$ values at T = 298K

$\log A$	$\Delta S^{\ddagger}$
11.2	−38 J/mole·K
12.2	−19
13.2	near 0
14.2	19
15.2	38